Natnael Alemseged Astaw

Applied AI Engineer | Agent Systems & LLM Evaluation

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Portfolio | LinkedIn | GitHub | Upwork

PROFILE

Applied AI engineer with 4+ years of software delivery experience building production agentic systems, domain evaluations, fine-tuning workflows, RAG pipelines, and LLM-powered APIs. Combines Python and TypeScript engineering with customer discovery, rapid prototyping, deployment, observability, and measurable iteration. Experienced with LangGraph, LLM-as-a-judge, preference optimization, FastAPI, AWS, and full-stack product integration.

EXPERIENCE

Senior Software Engineer
Jul 2025 - Present

Hire Armada | United States (Remote)

- Engineer production agentic backends and full-stack workflows using FastAPI, Node.js, LangGraph, MCP, Composio, Prisma, and vector memory.
- Delivered an agentic collaboration workspace with real-time events, email automation, semantic search, and tagged agents, improving team productivity by 20%.
- Built a social discovery platform backend supporting 10K+ daily active users, real-time matching and messaging, and 99.9% uptime.
- Shipped Gemini-powered chat and personalization features that increased retention by 35%, translating ambiguous product goals into observable production behavior.

AI Engineer
Jan 2025 - Aug 2025

DataCore Software | United States (Remote, Contract)

- Built FastAPI services for real-time detection, recognition, transcription, summarization, and translation using self-hosted LLMs.
- Deployed containerized inference workloads to AWS EKS with sub-200 ms inference and 99.9% uptime at peak load.
- Fine-tuned open models and optimized GPU inference with TensorRT and ECR images, reducing cost by 12%.
- Automated infrastructure and delivery with Terraform and GitHub Actions, including monitoring and rollback paths.

Senior Full-Stack Developer
Jan 2025 - May 2026

Startup Agile | United States (Remote)

- Converted product requirements into deployed Next.js, React Native, Expo, and API capabilities while completing milestones 15% ahead of schedule.
- Integrated AI assistants, analytics, notifications, maps, and secure data workflows across web and mobile products.
- Standardized TypeScript, tRPC, Prisma, Playwright, and Detox workflows, reducing production defects by 65%.

Senior Mobile Developer
Aug 2024 - Sep 2025

Qemer Software Technology PLC | Addis Ababa, Ethiopia (On-site)

- Led delivery of 10+ production applications, accelerating release cycles by 40% and strengthening direct product ownership.
- Mentored engineers and established reusable architecture patterns, reducing onboarding time by 30%.

EARLIER EXPERIENCE

ERP Functional Consultant **Mar 2023 - Sep 2024**
Red Cloud ICT Solutions | Addis Ababa, Ethiopia

- Progressed from Junior Consultant to Functional Consultant Assistant, supporting ERP discovery, configuration, testing, user enablement, and implementation delivery.

Freelance Web Developer **Aug 2022 - Mar 2023**
Independent | Remote

- Designed, built, and delivered responsive web solutions directly with clients from requirements through launch.

SELECTED INDEPENDENT DELIVERY

Sales-Domain LLM Evaluation Bench: Built a 240-task benchmark with deterministic checks, preference data, and a trained judge that achieved a +76.6pp held-out lift.

AI Chatbot Engineering: Completed a focused production chatbot engagement in Jan 2026; received a verified 5.0 Upwork rating.

Agentic Codebase Evaluation: Designed a LangGraph evidence workflow with independent judges, deterministic veto rules, and hallucination auditing.

Upwork Top Rated | 100% Job Success | Verified 5-star client outcomes

TECHNICAL SKILLS

Agents & Evaluation: LangGraph, agent orchestration, LLM-as-a-judge, domain benchmarks, deterministic evaluators, preference data, statistical testing

Fine-tuning & Inference: LoRA, DPO/SimPO, self-hosted LLMs, structured generation, sampling analysis, TensorRT, GPU inference optimization

Retrieval & Tool Use: RAG, MCP, Composio, vector databases, semantic search, memory systems, prompt engineering

Engineering: Python, FastAPI, TypeScript, Node.js, Next.js, PostgreSQL, Redis, Supabase, REST APIs, Socket.io

Experimentation & Delivery: Jupyter Notebook, Google Colab, AWS, Docker, Kubernetes/EKS, Terraform, GitHub Actions, Prometheus, Grafana

SELECTED TECHNICAL WRITING

When Generic Benchmarks Fail: Building a Sales-Domain Evaluation Bench from Scratch
DEV Community | May 2026

Why Pairing Your Bootstrap Is Necessary, And When It Stops Helping
DEV Community | May 2026

CERTIFICATIONS

AI Agent Engineering, AI Evaluation & Forward-Deployed Engineering
10 Academy | Issued May 2026

AWS Certified Cloud Practitioner
AWS | Oct 2023 - Feb 2027

CCNA: Introduction to Networks
Cisco | Issued Nov 2021

AWS Certified Solutions Architect - Associate
AWS | Feb 2024 - Feb 2027

Founders Academy
ALX Africa | Issued Jul 2024

EDUCATION

Bachelor of Science in Information Systems **Jan 2018 - Dec 2022**
Hawassa University | GPA: 3.57